

Interband Absorption in Gold from Fermi's Golden Rule: a first-principles rederivation of Rosei's ε_2 at the X and L critical points, in Rosei's own notation

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Abstract

This document rederives, from first principles and with every prefactor tracked, the Rosei-model interband $\varepsilon_2(\hbar\omega, T)$ of gold at the X and L critical points [Guerrisi, Rosei & Winsemius, Phys. Rev. B **12**, 557 (1975); hereafter simply “Rosei”], written throughout in Rosei's own notation, so that the paper's ten numbered equations emerge explicitly from one derivation. The route runs from Fermi's Golden Rule (FGR) for Bloch states in $\mathbf{k}$ -space, through the reduction of the transition sum to a one-dimensional energy integral by the change of variables $(u, v) = (k_\perp^2, k_\parallel^2)$, to the resolution of the double δ -constraint by a single 2×2 Jacobian. Seven of the ten equations are recreated verbatim; the printed Eqs. (4), (6) and (8) do not come out as written, and for each the failure is stated in one line and the corrected form derived (Sec. 5.1). The photonic density of states is derived by field quantization, the spontaneous-emission counterpart is assembled, and the pair is shown to close exactly into the detailed-balance relation between emission and absorption (van Roosbroeck–Shockley/Kirchhoff) — the global consistency check of all prefactors. A final section gives the $k_\parallel$ quadrature scheme that evaluates the model without touching its edge singularities: the groundwork for the numerical implementation.

Contents

1	Scope & sources	2
2	Fermi's Golden Rule in $\mathbf{k}$-space	2
2.1	Minimal coupling	2
2.2	Quantized field: absorption, stimulated and spontaneous emission	3
2.3	Bloch states and vertical transitions	3
2.4	Golden-rule rate, ε_2 , and the origin of the $1/\omega^2$	3
2.5	Occupation factor: Rosei's $[1 - f]$	4
3	Reduction of the $\mathbf{k}$-integral: the common machinery	4
3.1	The Rosei two-band model	4
3.2	Sorting transitions by final energy	5
3.3	Change of variables $(u, v) = (k_\perp^2, k_\parallel^2)$	5
3.4	The EDJDOS: general derivation for both X and L	5
3.5	The thermally weighted JDOS: Rosei's Eq. (7)	6
4	The two critical points, side by side	6
4.1	The X mass fork: sign of $\bar{B}_X$	8
5	Assembled ε_2	9
5.1	Discrepancies & suggested corrections	10
5.2	The Drude term: completing ε_2	11

6 Photonic density of states and the emission counterpart	12
6.1 Mode counting	12
6.2 Spontaneous emission rate of one pair	12
6.3 The interband PL spectrum and the Kirchhoff check	12
7 Numerics: evaluate in $k_{\parallel}$, not in E	13
A Conventions	14
B Dictionary of conventions across the repository	15
C Dimensional audit (Gaussian)	15

1 Scope & sources

Everything below is self-contained: it is built from Fermi’s Golden Rule and four primary sources.

- **Rosei 1975** (Guerrisi, Rosei & Winsemius; “Rosei” from here on) — the paper recreated. Only the X point is actually derived there, and the printed Eqs. (4), (6), (8) have defects — fixed in Sec. 5.1.
- **Rosei 1974** — the EDJDOS recipe and the L -point machinery (“will not be repeated here”; it is repeated here).
- **Christensen & Seraphin 1971** — the band structure: levels and masses. Rosei’s own source as well (his Ref. 4).
- **Johnson & Christy 1972** — the optical data the model is fitted to.

Notation is Rosei’s: bands are u (upper) and l (lower), all levels and masses are positive numbers with every sign written explicitly, and an equation tagged (n) is Rosei’s Eq. (n) recreated — primes, as in $(6')$, mark corrected forms. The rest of the conventions live in App. A.

Route map — shaded rows are the heart of the document, the Rosei recreation; the others are scaffolding that keeps it self-contained:

where	what happens	leans on
Sec. 2	golden-rule machinery, every constant tracked	—
Sec. 3	$\mathbf{k}$ -space reduction $\rightarrow$ Eqs. (4’), (5), (7)	Rosei 1975/1974
Sec. 4	X and L , side by side $\rightarrow$ Eqs. (6’), (8’)	Rosei 1975/1974
Sec. 5	ε_2 assembled $\rightarrow$ (9)–(11); scorecard; corrections; Drude	C&S; J&C
Sec. 6	photonic DOS, emission, detailed balance	Novotny–Hecht
Sec. 7	$k_{\parallel}$ quadrature for the implementation	—

2 Fermi’s Golden Rule in $\mathbf{k}$ -space

2.1 Minimal coupling

With the radiation field described by a vector potential $\mathbf{A}$ in the Coulomb gauge ($\nabla \cdot \mathbf{A} = 0$), the one-electron Hamiltonian is

$$\hat{H} = \frac{[\hat{\mathbf{p}} + \frac{e}{c}\mathbf{A}(\hat{\mathbf{r}}, t)]^2}{2m} + V(\hat{\mathbf{r}}) = \underbrace{\frac{\hat{p}^2}{2m} + V}_{\hat{H}_0} + \underbrace{\frac{e}{2mc}(\mathbf{A} \cdot \hat{\mathbf{p}} + \hat{\mathbf{p}} \cdot \mathbf{A})}_{\hat{H}_{\text{int}}} + \underbrace{\frac{e^2 A^2}{2mc^2}}_{\text{dropped}}. \quad (2.1)$$

In the Coulomb gauge $\hat{\mathbf{p}} \cdot \mathbf{A} = \mathbf{A} \cdot \hat{\mathbf{p}}$, so

$$\boxed{\hat{H}_{\text{int}} = \frac{e}{mc} \mathbf{A} \cdot \hat{\mathbf{p}}} \quad (2.2)$$

(the $\frac{1}{2}$ of the symmetrized form is consumed by the identity of the two terms). The A^2 term does not connect different bands to first order and is dropped.

2.2 Quantized field: absorption, stimulated and spontaneous emission

Expanding the field in a quantization volume V ,

$$\mathbf{A}(\mathbf{r}) = \sum_{\mathbf{q}, \lambda} \sqrt{\frac{2\pi\hbar c^2}{\omega_q V}} \hat{\mathbf{e}}_\lambda \left(\hat{a}_{\mathbf{q}\lambda} e^{i\mathbf{q} \cdot \mathbf{r}} + \hat{a}_{\mathbf{q}\lambda}^\dagger e^{-i\mathbf{q} \cdot \mathbf{r}} \right), \quad (2.3)$$

the matrix elements of $\hat{H}_{\text{int}}$ between joint electron–photon states carry the boson factors

$$|\langle n_q - 1 | \hat{a} | n_q \rangle|^2 = n_q \quad (\text{absorption}), \quad |\langle n_q + 1 | \hat{a}^\dagger | n_q \rangle|^2 = n_q + 1 \quad (\text{emission}). \quad (2.4)$$

The “+1” is the spontaneous channel: it survives at $n_q = 0$ and is invisible to a purely classical $\mathbf{A}$. The semiclassical golden rule gives absorption and *stimulated* emission only; spontaneous emission requires (2.4), and its mode sum produces the photonic density of states in Sec. 6.

2.3 Bloch states and vertical transitions

For Bloch states $\psi_{n\mathbf{k}} = e^{i\mathbf{k} \cdot \mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}) / \sqrt{V}$, the matrix element of $e^{\pm i\mathbf{q} \cdot \mathbf{r}} \hat{\mathbf{e}} \cdot \hat{\mathbf{p}}$ enforces $\mathbf{k}' = \mathbf{k} \pm \mathbf{q}$. At optical frequencies $q \sim \omega/c \sim 10^{-3} \text{ \AA}^{-1}$ is negligible on the zone scale, so transitions are vertical ($\mathbf{k}' = \mathbf{k}$) and

$$M_{ul}(\mathbf{k}) \equiv \langle u\mathbf{k} | \hat{\mathbf{e}} \cdot \hat{\mathbf{p}} | l\mathbf{k} \rangle, \quad |\langle f | \hat{H}_{\text{int}} | i \rangle|^2 = \left(\frac{e}{mc} \right)^2 \frac{2\pi\hbar c^2}{\omega V} |M_{ul}(\mathbf{k})|^2 \times \{n_q \text{ or } n_q + 1\}. \quad (2.5)$$

2.4 Golden-rule rate, ε_2 , and the origin of the $1/\omega^2$

FGR for one photon mode, then summing over $\mathbf{k}$ (factor 2 for spin, $\sum_{\mathbf{k}} \rightarrow V \int d^3k / (2\pi)^3$) with Pauli blocking, gives the photon absorption rate. Equating the absorbed power to the electromagnetic dissipation rate $P_{\text{abs}} = \frac{1}{8\pi} \omega \varepsilon_2(\omega) |\mathbf{E}_0|^2 V$ of a classical field $\mathbf{E} = \mathbf{E}_0 \cos \omega t$ yields

$$\boxed{\varepsilon_2(\omega) = \frac{4\pi^2 e^2}{m^2 \omega^2} \frac{2}{(2\pi)^3} \int_{\text{BZ}} d^3k |M_{ul}(\mathbf{k})|^2 \delta(\hbar\omega_u(\mathbf{k}) - \hbar\omega_l(\mathbf{k}) - \hbar\omega) [1 - f(\hbar\omega_u(\mathbf{k}))]} \quad (2.6)$$

per pair of bands (SI: replace $4\pi e^2 \rightarrow e^2/\varepsilon_0$). The occupation factor is Rosei’s, discussed in Sec. 2.5.

Where does the $1/\omega^2$ up front come from? It is fixed before any band structure enters — and it is worth watching, because emission will carry a *different* power (Sec. 6). Unpack (2.6) into its three ω -carrying pieces, marking the net power each one contributes:

$$\varepsilon_2 = \underbrace{\frac{8\pi P_{\text{abs}}/V}{\omega}}_{-1} \frac{E_0^2}{E_0^2}, \quad P_{\text{abs}} = \underbrace{\hbar\omega}_{+1} \times \text{rate}, \quad \text{rate} \propto \left(\frac{eA_0}{mc} \right)^2 = \frac{e^2 E_0^2}{m^2} \underbrace{\frac{1}{\omega^2}}_{-2} \quad (A_0 = cE_0/\omega). \quad (2.7)$$

The +1 (one quantum per event) cancels the −1 (the ω in the definition of dissipation); the −2 of the vertex is all that survives. It is the price of writing the $\mathbf{A} \cdot \mathbf{p}$ coupling in terms of the measured field — pure kinematics, no density of states involved. (Length-gauge cross-check: $|\mathbf{p}_{ul}|^2 = m^2 \omega_{ul}^2 |\mathbf{r}_{ul}|^2$ hides the same factor inside the dipole element, so on shell the two forms agree identically.)

Near a critical point the interband matrix element varies slowly; following Rosei we freeze it and average over polarization, in his gradient convention:

$$|P|^2 \equiv |\langle u | \nabla | l \rangle|^2 \quad (\text{length}^{-2}), \quad \overline{|M_{ul}|^2} = \overline{|\hat{\mathbf{e}} \cdot \mathbf{p}_{ul}|^2} = \frac{\hbar^2 |P|^2}{3}. \quad (2.8)$$

Per critical-point family this turns (2.6) into the working form

$$\varepsilon_2(\hbar\omega, T) = \frac{4\pi^2 e^2 \hbar^2}{3m^2 \omega^2} \cdot \frac{2}{(2\pi)^3} N |P|^2 \int_{\text{half-nbhd}} d^3k \delta(\mathcal{E}_{lu}(\mathbf{k}) - \hbar\omega) [1 - f(E)] \Big|_{E=\hbar\omega_u(\mathbf{k})}, \quad (2.9)$$

with the interband energy $\mathcal{E}_{lu} \equiv \hbar\omega_u - \hbar\omega_l$ (Sec. 3.1 connects it to Rosei's Ω_{lu}) and N the half-neighborhood count of App. A.

2.5 Occupation factor: Rosei's $[1 - f]$

The exact net weight of (2.6) — absorption minus stimulated emission — is the difference $f(E - \hbar\omega) - f(E)$ read at the final-state energy $E = \hbar\omega_u(\mathbf{k})$. For the noble-metal d bands the initial state sits at $E - \hbar\omega \lesssim -1.5 \text{ eV} \ll -k_B T$, so $f(E - \hbar\omega) = 1$ to machine precision and the difference *is* Rosei's factor,

$$f(E - \hbar\omega) - f(E) \longrightarrow [1 - f(E, T)], \quad (2.10)$$

which is used throughout this document, exactly as Rosei uses it. (The *emission* weight is a different object — a product, not a difference — and is derived in Sec. 6.)

3 Reduction of the $\mathbf{k}$ -integral: the common machinery

3.1 The Rosei two-band model

Both critical points are axially symmetric parabolic pairs. The parabolic dispersion relations are the primary sources': at X they are Rosei's own Eqs. (1)–(2); at L they are the forms of the 1974 Ag analysis, which the 1975 paper adopts for gold; levels and masses for both come from the relativistic band calculation of Christensen & Seraphin — Rosei's source too. Why label the bands by E and $E - \hbar\omega$? Because a vertical transition at photon energy $\hbar\omega$ connects exactly those two energies, so the occupation factors can be read off directly at the final energy E . With the positive levels/masses of App. A, generic levels $\hbar\omega_{u0} > 0$ (empty, above E_F) and $\hbar\omega_{l0} > 0$ (occupied d level below E_F), and the *topology switch* s ,

$$\boxed{\begin{aligned} E = \hbar\omega_u(\mathbf{k}) &= +\hbar\omega_{u0} + A_u k_\perp^2 + s B_u k_\parallel^2, & s_X &= -1 \quad (\text{saddle}), \\ E - \hbar\omega &= \hbar\omega_l(\mathbf{k}) = -\hbar\omega_{l0} - A_l k_\perp^2 - B_l k_\parallel^2, & s_L &= +1 \quad (\text{minimum}), \end{aligned}} \quad (3.1)$$

where $(\hbar\omega_{u0}, \hbar\omega_{l0}) = (\hbar\omega_{X_6^-}, \hbar\omega_{X_7^+})$ at X and $(\hbar\omega_{L_6^-}, \hbar\omega_{L^+})$ at L ; the lower band is a local maximum in both directions at both points. The interband energy is then

$$\mathcal{E}_{lu}(\mathbf{k}) = \mathcal{E}_g + \bar{A} k_\perp^2 + \bar{B} k_\parallel^2, \quad \mathcal{E}_g = \hbar\omega_{u0} + \hbar\omega_{l0}, \quad \bar{A} = A_u + A_l > 0, \quad \boxed{\bar{B} = B_l + s B_u} \quad (3.2)$$

— the entire topological difference between X and L is carried by s through $\bar{B}$ (and the determinant $\mathcal{D}$ below).

A note on reach: (3.1) are local expansions, and the constant matrix elements behind Eq. (9) are claimed by Rosei only in a region $\sim(\pi/10a)^3$ around each point — yet no explicit k_{max} will appear in any integration bound below. That is Rosei's own bookkeeping, not an omission: what polices locality instead (the photon-energy range) is quantified after the window table of Sec. 5.

A note on symbols: Rosei's own Ω_{lu} is not this energy but the *constraint* built from it,

$$\Omega_{lu}(\mathbf{k}) \equiv \hbar\omega_u - \hbar\omega_l - \hbar\omega = \mathcal{E}_{lu}(\mathbf{k}) - \hbar\omega = 0, \quad (3)$$

whose zero set is the constant-energy-difference surface (CEDS) — the surface on which everything below happens. We keep the two apart: $\mathcal{E}_{lu}$ is a function on $\mathbf{k}$ -space; Rosei's Eq. (3), $\Omega_{lu} = 0$, picks out the surface. The gold parameters are collected in the tables of Sec. 5.

3.2 Sorting transitions by final energy

The occupation in (2.9) depends on $\mathbf{k}$ only through the final energy $E = \hbar\omega_u(\mathbf{k})$ — that is where temperature enters — so before any geometry we *sort the transitions by E* , by slipping the identity $1 = \int dE \delta(\hbar\omega_u(\mathbf{k}) - E)$ into (2.9):

$$\begin{aligned} \int_{k_{\parallel} > 0} d^3k \delta(\mathcal{E}_{lu} - \hbar\omega) [1 - f] &= \int_{E_{\min}}^{E_{\max}} dE [1 - f(E)] \underbrace{\int_{k_{\parallel} > 0} d^3k \delta(\hbar\omega_u - E) \delta(\mathcal{E}_{lu} - \hbar\omega)}_{(2\pi)^3 D_{l \rightarrow u}(E, \hbar\omega)} \\ &\equiv (2\pi)^3 \mathcal{J}_{l \rightarrow u}(\hbar\omega, T). \end{aligned} \quad (3.3)$$

The limits $[E_{\min}, E_{\max}]$ are simply where the double- δ has support; they are worked out per critical point in Sec. 4. Read (3.3) from the outside in: the outer object $\mathcal{J}_{l \rightarrow u}$ is already Rosei's thermally weighted JDOS — his Eq. (7), waiting only for its integrand — and that integrand, the inner integral, is his central object: the *energy-distributed joint density of states* (EDJDOS), transitions per unit crystal volume, per unit final energy, per unit transition energy (one half-neighborhood, one spin). Both δ 's are accounted for: energy conservation was already in (2.9); the second is the inserted identity, and it is what lets $\mathcal{J}$ weight each E by its own occupation.

3.3 Change of variables $(u, v) = (k_{\perp}^2, k_{\parallel}^2)$

This is the step where the present derivation simplifies Rosei's geometric construction — and the form of (3.3) itself dictates it. The two δ -functions in (3.3) have arguments $\hbar\omega_u(\mathbf{k}) - E$ and $\mathcal{E}_{lu}(\mathbf{k}) - \hbar\omega$, and by (3.1) and (3.2) both are *affine in the squared coordinates* $k_{\perp}^2$ and $k_{\parallel}^2$. A δ -function of a linear argument collapses trivially; so choose the variables in which the constraints *are* linear. On the quadrant $u, v \geq 0$ with $(u, v) = (k_{\perp}^2, k_{\parallel}^2)$, fixing the final energy E and the interband energy $\mathcal{E}_{lu} = \hbar\omega$ reads

$$\begin{pmatrix} E - \hbar\omega_{u0} \\ \hbar\omega - \mathcal{E}_g \end{pmatrix} = \underbrace{\begin{pmatrix} A_u & s B_u \\ \bar{A} & \bar{B} \end{pmatrix}}_M \begin{pmatrix} u \\ v \end{pmatrix}, \quad \mathcal{D} \equiv \det M = A_u B_l - s A_l B_u. \quad (3.4)$$

Stacking the two constraints *is* the construction of the transition matrix M ; its determinant is the Jacobian that will collapse the two δ -functions below. Inverting,

$$\boxed{u = \frac{\bar{B}(E - \hbar\omega_{u0}) - s B_u(\hbar\omega - \mathcal{E}_g)}{\mathcal{D}}, \quad v = k_{\parallel}^2 = \frac{A_u(\hbar\omega - \mathcal{E}_g) - \bar{A}(E - \hbar\omega_{u0})}{\mathcal{D}}} \quad (3.5)$$

and the physical window is exactly $\{u \geq 0\} \cap \{v \geq 0\}$. The two point-specific determinants:

$$\mathcal{D}_X = A_u B_l + A_l B_u > 0 \text{ always (either sign of } \bar{B}_X), \quad \mathcal{D}_L = A_u B_l - A_l B_u (< 0 \text{ for Au}). \quad (3.6)$$

3.4 The EDJDOS: general derivation for both X and L

Now reduce the inner integral of (3.3) all the way — this is where $\mathcal{D}$ and $\mathcal{F}$ come from. Axial symmetry gives $d^3k = 2\pi k_{\perp} dk_{\perp} dk_{\parallel}$; in the squared variables $k_{\perp} dk_{\perp} = \frac{1}{2} du$, $dk_{\parallel} = dv/(2\sqrt{v})$:

$$\frac{1}{(2\pi)^3} \int_{k_{\parallel} > 0} d^3k = \frac{1}{(2\pi)^3} \frac{\pi}{2} \int_0^{\infty} du \int_0^{\infty} \frac{dv}{\sqrt{v}} = \frac{1}{16\pi^2} \int_0^{\infty} du \int_0^{\infty} \frac{dv}{\sqrt{v}}. \quad (3.7)$$

The two δ 's are exactly the two rows of M in (3.4); linear δ 's collapse onto the unique solution (3.5) divided by $|\det M| = |\mathcal{D}|$, and $\sqrt{v_*} = k_{\parallel}(E, \hbar\omega)$:

$$D_{l \rightarrow u}(E, \hbar\omega) = \frac{1}{16\pi^2 |\mathcal{D}|} \frac{1}{k_{\parallel}(E, \hbar\omega)}. \quad (3.8)$$

This is the whole result of the reduction: it contains only $\mathcal{D}$ and $k_{\parallel}$ — no $\mathcal{F}$ yet. Rosei's $\mathcal{F}_{l \rightarrow u}$ is not an independent input but a repackaging of $\mathcal{D}$, manufactured here. Expanding $\mathcal{D}_X$ in the masses,

$$\mathcal{D}_X = A_u B_l + A_l B_u = \left(\frac{\hbar^2}{2}\right)^2 \frac{m_{l\perp} m_{u\parallel} + m_{l\parallel} m_{u\perp}}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}}, \quad (3.9)$$

and defining $\mathcal{F}_{l \rightarrow u} \equiv \hbar^2/(2\sqrt{|\mathcal{D}|})$ as pure shorthand, the $\hbar^2/2$ cancels and Rosei's Eq. (5) falls out — derived, not assumed — together with its L analogue:

$$\mathcal{F}_X = \left[\frac{m_{l\perp} m_{u\parallel} + m_{l\parallel} m_{u\perp}}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}} \right]^{-1/2}, \quad \mathcal{F}_L = \left[\frac{|m_{l\perp} m_{u\parallel} - m_{l\parallel} m_{u\perp}|}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}} \right]^{-1/2} \quad (5)$$

(the sum/difference structure is the $s = \mp 1$ dichotomy of (3.6)). Reading the shorthand backwards, $|\mathcal{D}| = \hbar^4/4\mathcal{F}^2$, and (3.8) becomes the EDJDOS in Rosei's own symbols:

$$D_{l \rightarrow u}(E, \hbar\omega) = \frac{1}{16\pi^2 |\mathcal{D}|} \frac{1}{k_{\parallel}(E, \hbar\omega)} = \frac{\mathcal{F}_{l \rightarrow u}^2}{4\pi^2 \hbar^4} \frac{1}{k_{\parallel}(E, \hbar\omega)} \quad (4')$$

supported on $u, v \geq 0$, with $k_{\parallel}$ from (3.5).

Where the printed (4) falls. Rosei prints $D_{l \rightarrow u} = (8\pi^2 \hbar^2)^{-1} \mathcal{F}/k_{\parallel}$. With $|P|^2$ the squared gradient matrix element of (2.8), the pair {printed (4), printed (9)} does not balance dimensionally, whereas (4') — forced, with nothing left to choose, by the definition of $D_{l \rightarrow u}$ in (3.3) — closes the FGR chain exactly through to ε_2 (Sec. 5). The two forms differ only by the constant $2\mathcal{F}/\hbar^2$ per critical point; a constant cannot bend a line shape, and it is absorbed into the fitted strength $S = \mathcal{F}|P|^2$ of Eq. (10) — which is why the slip is invisible in the published numbers. Details in Sec. 5.1.

3.5 The thermally weighted JDOS: Rosei's Eq. (7)

With $D_{l \rightarrow u}$ in hand, the outer integral of (3.3) is, verbatim as Rosei writes it,

$$\mathcal{J}_{l \rightarrow u}(\hbar\omega, T) = \int_{E_{\min}}^{E_{\max}} D_{l \rightarrow u}(E, \hbar\omega) [1 - f(E, T)] dE \quad (7)$$

with the windows $[E_{\min}, E_{\max}]$ set by the geometry ($u, v \geq 0$) and, where the geometry leaves an edge open, by the occupation itself — next section.

4 The two critical points, side by side

Everything so far is common. The topology forks the derivation through two signs only: s (shape of $\bar{B}$) and $\text{sign}(\mathcal{D})$ (which window edge is singular). The X column follows Rosei's own branch, $\bar{B}_X < 0$ (see the mass fork, Sec. 4.1); his equations then pop out directly.

X point ($s = -1$)

Bands

$$E = +\hbar\omega_{X_6^-} + \frac{\hbar^2 k_\perp^2}{2m_{u\perp}} - \frac{\hbar^2 k_\parallel^2}{2m_{u\parallel}},$$

$$E - \hbar\omega = -\hbar\omega_{X_7^+} - \frac{\hbar^2 k_\perp^2}{2m_{l\perp}} - \frac{\hbar^2 k_\parallel^2}{2m_{l\parallel}}.$$

Rosei's Eqs. (1)–(2) verbatim: the X_6^- (sp) band curves up in the face, down along Δ — a saddle; X_7^+ (d) is a maximum.

Transition surface (CEDS)

$$\mathcal{E}_{lu} = \mathcal{E}_g^X + \bar{A}_X k_\perp^2 + \bar{B}_X k_\parallel^2, \quad \bar{B}_X = B_l - B_u < 0$$

on Rosei's branch ($m_{u\parallel} < m_{l\parallel}$: the sp band falls along Δ faster than the d band). The CEDS is an *open* hyperboloid at every $\hbar\omega$: transitions exist on both sides of $\mathcal{E}_g^X$ — a real sub-gap tail.

Determinant

$$\mathcal{D}_X = A_u B_l + A_l B_u > 0 \quad (\text{either branch}).$$

Constrained $k_\parallel$

$$k_\parallel^2 = \frac{2\mathcal{F}_X^2}{\hbar^2} \left[\frac{\hbar\omega - \hbar\omega_{X_7^+} - E}{m_{u\perp}} + \frac{\hbar\omega_{X_6^-} - E}{m_{l\perp}} \right].$$

Rosei's Eq. (6), corrected — called (6') throughout; see Sec. 5.1.

Window

E decreases without bound along the CEDS ($dE \propto -\mathcal{D}_X dv < 0$): no geometric floor. The integral is cut by occupation alone — Rosei's practical floor $E_{\min} = -20k_B T$. The ceiling is geometric and changes character at the gap:

$$E_{\max} = \hbar\omega_{X_6^-} + \begin{cases} \frac{A_u}{\bar{A}_X}(\hbar\omega - \mathcal{E}_g^X), & \hbar\omega \geq \mathcal{E}_g^X, \\ -\frac{B_u}{|\bar{B}_X|}(\mathcal{E}_g^X - \hbar\omega), & \hbar\omega < \mathcal{E}_g^X, \end{cases}$$

the lower line being Rosei's Eq. (8), corrected — (8'), Sec. 5.1.

L point ($s = +1$)

Bands

$$E = +\hbar\omega_{L_6^-} + \frac{\hbar^2 k_\perp^2}{2m_{u\perp}} + \frac{\hbar^2 k_\parallel^2}{2m_{u\parallel}},$$

$$E - \hbar\omega = -\hbar\omega_{L^+} - \frac{\hbar^2 k_\perp^2}{2m_{l\perp}} - \frac{\hbar^2 k_\parallel^2}{2m_{l\parallel}}.$$

Not derived in the 1975 paper (borrowed from Rosei 1974); rederived here with the same tools. L_6^- is a *minimum* in both directions; L^+ (d top) a maximum. The only change from X is the sign of the upper $k_\parallel^2$ term; that one flip cascades through everything below.

Transition surface (CEDS)

$$\mathcal{E}_{lu} = \mathcal{E}_g^L + \bar{A}_L k_\perp^2 + \bar{B}_L k_\parallel^2, \quad \bar{B}_L = B_l + B_u > 0$$

unconditionally: a closed ellipsoid, existing only for $\hbar\omega \geq \mathcal{E}_g^L$. *No sub-gap tail at L* — the sharp-vs-tailed contrast of the two edges is baked in right here.

Determinant

$$\mathcal{D}_L = A_u B_l - A_l B_u < 0 \quad \text{for Au}$$

$$(m_{l\perp} m_{u\parallel} = 0.084 < m_{l\parallel} m_{u\perp} = 0.247).$$

Constrained $k_\parallel$

$$k_\parallel^2 = \frac{2\mathcal{F}_L^2}{\hbar^2} \left[\frac{E - (\hbar\omega - \hbar\omega_{L^+})}{m_{u\perp}} + \frac{E - \hbar\omega_{L_6^-}}{m_{l\perp}} \right].$$

Same inversion, $\mathcal{D}_L < 0$ reversing both inequalities.

Window

Both edges geometric ($v \geq 0$ now gives the *lower* edge, $u \geq 0$ the upper):

$$E_{\min} = \hbar\omega_{L_6^-} + \frac{A_u}{\bar{A}_L}(\hbar\omega - \mathcal{E}_g^L),$$

$$E_{\max} = \hbar\omega_{L^+} + \frac{B_u}{\bar{B}_L}(\hbar\omega - \mathcal{E}_g^L),$$

slopes 0.745 and 0.896 for Au. The whole window rides upward from $\hbar\omega_{L_6^-} = 0.89$ eV (Remark 4.1).

Singular edge

Above the gap the ceiling is the $k_{\parallel} \rightarrow 0$ circle, where $D_{l \rightarrow u} \propto 1/k_{\parallel}$ diverges:

$$\frac{1}{k_{\parallel}(E)} = \sqrt{\frac{\mathcal{D}_X}{\bar{A}_X}} \frac{1}{\sqrt{E_{\max} - E}}.$$

Below the gap the ceiling is the finite $k_{\perp} = 0$ point: no divergence.

Onset

At $T = 0$ absorption starts when $E_{\max}$ crosses E_F :

$$\hbar\omega_{\text{on}} = \mathcal{E}_g^X - \hbar\omega_{X_6^-} \frac{m_{l\parallel} - m_{u\parallel}}{m_{l\parallel}} \simeq 1.83 \text{ eV},$$

the piezomodulation threshold; the tail 1.83–1.94 eV is purely geometric, and the onset is *step-like* (Rosei’s word), not divergent.

Singular edge

From (3.5) with $\mathcal{D}_L < 0$,

$$k_{\parallel}^2 = \frac{\bar{A}_L}{|\mathcal{D}_L|} (E - E_{\min}),$$

so $D_{l \rightarrow u} \propto 1/k_{\parallel}$ diverges at the *lower* limit:

$$\frac{1}{k_{\parallel}(E)} = \sqrt{\frac{|\mathcal{D}_L|}{\bar{A}_L}} \frac{1}{\sqrt{E - E_{\min}}}.$$

Onset

At $\hbar\omega = \mathcal{E}_g^L$ exactly, with $\mathcal{J}_L \propto \sqrt{\hbar\omega - \mathcal{E}_g^L}$; the inverse-square-root weight sits on the onset edge itself, which is why the L edge is sharp and strong.

Punchline. The same machinery, forked only by two signs, lands on two very different absorption edges. At L the $1/k_{\parallel}$ divergence of $D_{l \rightarrow u}$ sits on the *onset* edge itself, so absorption switches on sharply and strongly at $\mathcal{E}_g^L = 2.45$ eV. At X the divergence sits at the *top* of the window, far from onset; the onset itself is a soft, step-like rise near 1.83 eV with a geometric sub-gap tail. One interband edge that is really two — soft at X , sharp half an eV above it at L — and that is exactly the “splitting of the interband absorption edge” of the 1975 title.

Remark 4.1 (The L level and the Fermi-surface neck) *Rosei’s qualitative discussion attributes the strength of the L singularity to the onset CEDS being tangent to the Fermi surface along the neck — which requires L_6^- below E_F . The fitted scheme ($\mathcal{E}_g^L = 2.45$, $\hbar\omega_{L^+} = 1.56 \Rightarrow \hbar\omega_{L_6^-} = +0.89 \text{ eV}$) puts the whole L window above E_F , making the L channel inert through the Fermi factor. Both statements cannot hold in one parabolic model; the fitted scheme is what the code implements, and the tension is recorded here rather than papered over.*

4.1 The X mass fork: sign of $\bar{B}_X$

One assignment at X I am not sure of, flagged because it flips a sign. First, where the masses come from: they are not printed as numbers anywhere — they are read off the band diagram of C&S (Fig. 5), by fitting a parabola to the printed $E(\mathbf{k})$ curves around X , along Δ and in the face, and taking $m = \hbar^2(d^2E/dk^2)^{-1}$. Rosei’s masses come from exactly the same source (“the optical masses and the gaps... were taken from Ref. 4”, his Ref. 4 being C&S), so the two readings of the same figure ought to agree. They do not: my extraction gives $m_{u\parallel} = 0.40$, $m_{l\parallel} = 0.15$ — the *opposite* $\parallel$ ordering to Rosei’s, hence $\bar{B}_X > 0$. On that branch the sub-gap picture changes: the integration window becomes *closed* — finite at both ends by geometry, as at L — instead of Rosei’s bottomless, occupation-cut window; there is no geometric sub-gap tail, and the onset is a $\sqrt{\hbar\omega - \mathcal{E}_g^X}$ rise sitting exactly at the gap (any observed tail below 1.94 eV would then be broadening only, and the $-20k_B T$ floor would overshoot the true, geometric one).

Two facts keep the fork contained:

- Above the gap the branches are *indistinguishable*: same line shape, same singular edge. The $\parallel$ masses enter only through $\mathcal{F}_X$, and the fit lumps $\mathcal{F}_X$ into $S_X = \mathcal{F}_X |P_X|^2$ — so no fit to ε_2 data can tell the branches apart; only sub-gap behaviour or the extracted $|P_X|^2$ could. (The above-gap window slope $A_u/\bar{A}_X$ — 0.62 on Rosei’s assignment, 0.38 on mine — is set by the $\perp$ masses, a separate assignment.)
- Either ordering makes the d band the *lighter* one along Δ , which goes against the flat- d prior — so the likeliest culprit is a transcription swap in my own mass table, not new physics.

Bottom line: until C&S Fig. 5 is re-measured, this document follows Rosei’s branch ($\bar{B}_X < 0$) everywhere, and quotes the branch wherever it matters.

5 Assembled ε_2

Insert the EDJDOS (4′) into the FGR form (2.9). The bridge is the definition of $D_{l \rightarrow u}$ itself, Eq. (3.3): the $(2\pi)^3$ cancels, and with spin (the explicit 2) and the half-neighborhood counts $N_X = 6$, $N_L = 8$,

$$\varepsilon_2(\hbar\omega, T) = \frac{4\pi^2 e^2 \hbar^2}{3m^2 \omega^2} \cdot 2 \cdot \sum_{i=X,L} N_i |P_i|^2 \mathcal{J}_i(\hbar\omega, T), \quad (5.1)$$

which tidies to Rosei’s second headline equation, with no leftover constant:

$$\boxed{\varepsilon_2(\hbar\omega, T) = \frac{8\pi^2 e^2 \hbar^4}{3m^2 (\hbar\omega)^2} \sum_{i=X,L} N_i |P_i|^2 \mathcal{J}_i(\hbar\omega, T)} \quad (9)$$

(Rosei leaves the N_i implicit, so the fitted $|P_i|^2$ really means $N_i |P_i|^2$ up to normalization — harmless for the quoted ratio). That clean landing needs three things at once: $|P|^2$ is the squared gradient element (2.8); $D_{l \rightarrow u}$ is (4′), not the printed (4); spin and N_i are kept where shown. The quantities the data actually fix are the strengths

$$S_X = \mathcal{F}_X |P_X|^2, \quad S_L = \mathcal{F}_L |P_L|^2, \quad (10)$$

with $|P_X/P_L|^2 = 0.370$ from the fit to Johnson–Christy; under the corrected (4′) the fitted combination maps onto the printed S_i through exactly the constant $2\mathcal{F}_i/\hbar^2$ of Sec. 3.4 — every published number keeps its meaning.

Fully substituted — the explicit integral a computer evaluates — (4′) into (9), the $\hbar^4$ ’s cancelling:

$$\boxed{\varepsilon_2(\hbar\omega, T) = \frac{2e^2}{3m^2 (\hbar\omega)^2} \sum_{i=X,L} N_i \mathcal{F}_i^2 |P_i|^2 \int_{E_{\min}^{(i)}}^{E_{\max}^{(i)}} \frac{1 - f(E, T)}{k_{\parallel}^{(i)}(E, \hbar\omega)} dE} \quad (11)$$

with $k_{\parallel}^X$ from (6′), $k_{\parallel}^L$ from its L twin (same bracket, every slot sign-reversed), and windows:

	$E_{\min}$	$E_{\max}$	edge behaviour
$X, \hbar\omega \geq \mathcal{E}_g^X$	$-20k_B T$ (occupation)	$\hbar\omega_{X_6^-} + \frac{A_u}{A_X}(\hbar\omega - \mathcal{E}_g^X)$	$1/\sqrt{}$ at $E_{\max}$
$X, \hbar\omega < \mathcal{E}_g^X$	$-20k_B T$ (occupation)	sub-gap line of (8′)	finite everywhere
$L, \hbar\omega \geq \mathcal{E}_g^L$	$\hbar\omega_{L_6^-} + \frac{A_u}{A_L}(\hbar\omega - \mathcal{E}_g^L)$	$\hbar\omega_{L_6^-} + \frac{B_u}{B_L}(\hbar\omega - \mathcal{E}_g^L)$	$1/\sqrt{}$ at $E_{\min}$
$L, \hbar\omega < \mathcal{E}_g^L$	—	—	no states

Domain of validity — where is $k_{\max}$? The parabolic bands (3.1) and the constant $|P_i|^2$ hold only near the critical points; Rosei sizes the trusted region as $\sim(\pi/10a)^3$ (matrix-element constancy, his Sec. IV), while his Figs. 3 and 5 draw the CEDS out to $k \sim 2 \times \pi/4a$. Yet none of the windows

above carries an explicit $k_{\max}$ — they encode CEDS geometry and occupation only. This is faithful to the paper: Rosei polices locality through the *photon energy* instead, fitting only to 2.7 eV — “only limited regions in K space around X and around L are involved ... when the fitting is extended a few tenths of an eV above the absorption edge”, while “the danger of counting transitions twice and of a breakdown of our approximations” (both channels are band-5 $\rightarrow$ band-6) “may therefore arise at higher photon energies”. The numbers behind that policy, with the parameters below ($a_{\text{Au}} = 4.08 \text{ \AA}$, so $\pi/4a = 0.19 \text{ \AA}^{-1}$):

- *L is safely local*: the CEDS ellipsoid lies entirely inside $|\mathbf{k}| \leq \pi/4a$ for $\hbar\omega - \mathcal{E}_g^L \leq \bar{A}_L(\pi/4a)^2 \simeq 0.79 \text{ eV}$, i.e. up to $\hbar\omega \simeq 3.2 \text{ eV}$ — the whole fit range (inside $\pi/10a$ only for $\hbar\omega \lesssim 2.6 \text{ eV}$).
- *X is marginal by construction*: the occupied edge of the window ($E \simeq E_F$, where the Fermi step sits) lies at $|\mathbf{k}| \simeq (0.8\text{--}1.5) \times \pi/4a$ for $\hbar\omega = 2.0\text{--}2.7 \text{ eV}$ — several times the strict constant- $|P|^2$ radius, and exactly the scale of Rosei’s own figures. The $-20k_B T$ floor is an *occupation* cutoff, not a locality one — it maps to $|\mathbf{k}|$ up to $\sim 1.9 \times \pi/4a$ — but in equilibrium everything beyond the Fermi step carries $[1 - f] \approx 0$, so the occupation acts as the de facto k -cutoff. Nothing in the model *checks* this.

Operationally: trust the model where Rosei did, $\hbar\omega \lesssim 2.7 \text{ eV}$; above that, locality and single-counting fail together. Out of equilibrium the occupation excuse at X weakens — see the locality guard in Sec. 7.

Parameters (levels: Rosei’s fit; masses: C&S, the X row on Rosei’s branch, subject to Sec. 4.1; $\mathcal{F}$ in units of the free-electron mass m ; $\mathcal{D}$ in units $(\hbar^2/2m)^2$):

	$\mathcal{E}_g \text{ (eV)}$	upper level (eV)	lower level (eV)	$m_{u\perp}$	$m_{u\parallel}$	$m_{l\perp}$	$m_{l\parallel}$
X	1.94	$\hbar\omega_{X_6^-} = 0.17$	$\hbar\omega_{X_7^+} = 1.77$	0.19	0.15	0.31	0.40
L	2.45	$\hbar\omega_{L_6^-} = 0.89$	$\hbar\omega_{L^+} = 1.56$	0.24	0.12	0.70	1.03

	$\mathcal{D}$	$\mathcal{F}/m$	window slopes	singular edge
X	+34.7	0.170	$E_{\max}$ (above gap): 0.62	upper ($k_{\parallel} \rightarrow 0$)
L	-7.86	0.357	$E_{\min}$: 0.745, $E_{\max}$: 0.896	lower ($k_{\parallel} \rightarrow 0$)

Scorecard — the ten equations, one by one:

Rosei	here	status	issue as printed
(1)	(1)	recreated	—
(2)	(2)	recreated	—
(3)	(3)	recreated	—
(4)	(4')	suggested fix	prefactor $(8\pi^2\hbar^2)^{-1}\mathcal{F} \rightarrow (4\pi^2\hbar^4)^{-1}\mathcal{F}^2$; off by const. $2\mathcal{F}/\hbar^2$
(5)	(5)	recreated	—
(6)	(6')	suggested fix	dimensionally impossible as printed
(7)	(7)	recreated	—
(8)	(8')	suggested fix	$m_{u\parallel} \leftrightarrow m_{l\parallel}$ subscripts swapped
(9)	(9)	recreated	N_i made explicit
(10)	(10)	recreated	absorbs the (4) constant; see App. B

5.1 Discrepancies & suggested corrections

At issue: the printed Eqs. (4), (6) and (8) of the 1975 paper. None of the three comes out of the rederivation as printed — and none of the three failures touches a published number. Each failure stated simply, then fixed.

Eq. (4). Printed: $D_{l \rightarrow u} = (8\pi^2\hbar^2)^{-1} \mathcal{F}_{l \rightarrow u} k_{\parallel}^{-1}$. *Why it fails*: the definition of $D_{l \rightarrow u}$ (Eq. (3.3)) leaves nothing to choose — the reduction of Sec. 3.4 gives $(16\pi^2|\mathcal{D}|)^{-1}k_{\parallel}^{-1} = \mathcal{F}^2/(4\pi^2\hbar^4k_{\parallel})$, Eq. (4'),

and with the gradient convention for $|P|^2$ only (4') makes $\{(4),(9)\}$ dimensionally consistent and lands (9) with no leftover constant. The printed form is off by

$$\frac{\text{printed (4)}}{\text{correct (4')}} = \frac{2\mathcal{F}_{l \rightarrow u}}{\hbar^2}, \quad (5.2)$$

a pure constant per critical point — it cannot bend a line shape, and it is absorbed once and for all into the fitted $S = \mathcal{F}|P|^2$ of Eq. (10). That is why the slip is easy to miss and costs nothing physically; but the assembly only closes into (9) if (4') is used.

Eq. (6). Printed:

$$k_{\parallel} = \left(\hbar\omega - \hbar\omega_{X_7^+} + \frac{\hbar^2}{2m_{l\perp}}(\hbar\omega_{X_6^-} - E) - \frac{\hbar^2}{2m_{u\perp}}E \right)^{1/2}. \quad (6, \text{ as printed})$$

Why it fails: a wavevector is set equal to the square root of a sum of *energies*, and inside the bracket a bare energy is added to $(\hbar^2/2m) \times \text{energy}$ terms — mismatched units throughout; the coefficients read $\hbar^2/2m$ where $2m/\hbar^2$ is needed for the bracket to come out as length^{-2} . Three small fixes turn it into (6'): square the left-hand side; invert the coefficients and pull out the overall $2\mathcal{F}^2/\hbar^2$; restore the missing $1/m_{u\perp}$ on the first term. The three numerator slots of

$$k_{\parallel}^2 = \frac{2\mathcal{F}_X^2}{\hbar^2} \left[\frac{\hbar\omega - \hbar\omega_{X_7^+} - E}{m_{u\perp}} + \frac{\hbar\omega_{X_6^-} - E}{m_{l\perp}} \right] = \frac{A_u(\hbar\omega - \mathcal{E}_g^X) - \bar{A}_X(E - \hbar\omega_{X_6^-})}{\mathcal{D}_X} \quad (6')$$

then line up with the print term by term — confirmation that (6') is what was meant: a typo, but a disabling one, since the printed formula is unusable as written.

Eq. (8). Printed: $E_{\max} = \hbar\omega_{X_6^-} + (\mathcal{E}_g^X - \hbar\omega) m_{u\parallel} / (m_{l\parallel} - m_{u\parallel})$. Solving the model for the $k_{\perp} = 0$ edge gives

$$E_{\max} = \hbar\omega_{X_6^-} + (\mathcal{E}_g^X - \hbar\omega) \frac{m_{l\parallel}}{m_{u\parallel} - m_{l\parallel}}, \quad (8')$$

i.e. the printed equation has $m_{u\parallel}$ and $m_{l\parallel}$ interchanged — one consistent swap, specific to (8): the subscripts of (6) are *not* swapped, so it is not the same slip repeated. *The tell:* on Rosei's own branch $m_{u\parallel} < m_{l\parallel}$, the printed denominator is positive and sends $E_{\max}$ *above* $\hbar\omega_{X_6^-}$ as $\hbar\omega$ drops below the gap — but the sub-gap CEDS has no states up there. (8') puts the sub-gap edge sensibly *below* $\hbar\omega_{X_6^-}$, which is what makes the tail (and the 1.83 eV step onset) work. On the alternative branch of Sec. 4.1 ($\bar{B}_X > 0$), the same algebraic edge is instead the *lower* window limit.

5.2 The Drude term: completing ε_2

Everything above is interband. The measured ε_2 of gold also contains the free-carrier (intraband) response, which the Rosei model deliberately leaves out — so to compare with data, or to fit, it has to be added back. The Drude dielectric function $\varepsilon(\omega) = \varepsilon_{\infty} - \omega_p^2 / [\omega(\omega + i\gamma)]$ gives

$$\varepsilon_2^D(\omega) = \frac{\omega_p^2 \gamma}{\omega(\omega^2 + \gamma^2)} \xrightarrow{\omega \gg \gamma} \frac{\omega_p^2 \gamma}{\omega^3}, \quad (5.3)$$

and in the optical window the high-frequency limit is all that is needed: for Au, $\hbar\omega_p \simeq 9.0$ eV and $\hbar\gamma \simeq 0.07$ eV, so $\omega \gg \gamma$ holds everywhere above ~ 1 eV (the fit code implements exactly this A_D/ω^3 form, $A_D = \omega_p^2 \gamma \simeq 5.7$ eV³). The total is a plain sum of channels:

$$\boxed{\varepsilon_2(\hbar\omega, T) = \varepsilon_2^D(\omega) + \frac{8\pi^2 e^2 \hbar^4}{3m^2 (\hbar\omega)^2} \sum_{i=X,L} N_i |P_i|^2 \mathcal{J}_i(\hbar\omega, T)} \quad (5.4)$$

The Drude part falls as ω^{-3} and carries no interband structure: below the X onset it *is* ε_2 , through the edge region it is a smooth declining background, and the temperature dependence of the edge sits entirely in the $\mathcal{J}_i$.

6 Photonic density of states and the emission counterpart

6.1 Mode counting

Periodic quantization in volume V : two transverse polarizations, $\omega = cq$:

$$\rho(\omega) d\omega = \frac{2}{V} \frac{V}{(2\pi)^3} 4\pi q^2 dq \implies \boxed{\rho(\omega) = \frac{\omega^2}{\pi^2 c^3}} \quad (6.1)$$

modes per unit volume per unit angular frequency. In a transparent host of index n , $c \rightarrow c/n$; in a structured environment $\rho \rightarrow \rho(\mathbf{r}, \omega)$, the *local* density of states (Novotny & Hecht) — precisely the photonic factor of the PL factorization used in this thesis.

6.2 Spontaneous emission rate of one pair

FGR with the quantized field (2.3) at $n_q = 0$, summed over modes, using $\sum_\lambda \int d\Omega_q |\hat{\epsilon}_\lambda \cdot \mathbf{p}_{ul}|^2 = \frac{8\pi}{3} |\mathbf{p}_{ul}|^2$ and $|\mathbf{p}_{ul}|^2 = \hbar^2 |P|^2$:

$$\boxed{W_{\text{sp}}(\omega) = \frac{4e^2 \omega \hbar |P|^2}{3m^2 c^3} = \underbrace{\frac{4\pi^2 e^2 \hbar |P|^2}{3m^2 \omega}}_{\text{matter}} \times \underbrace{\frac{\omega^2}{\pi^2 c^3}}_{\rho(\omega)}} \quad (6.2)$$

— the vacuum Einstein- A rate in the $\mathbf{A} \cdot \mathbf{p}$ form. The factorization into a matter part times $\rho(\omega)$ is exact and is the microscopic origin of the LDOS $\times$ electronic split assumed in Chapter 2.

Why $1/\omega$ in the matter part here, when the absorption prefactor of (2.6) carries $1/\omega^2$? Both rates contain the identical vertex $(e/mc)^2 |A|^2 |M_{ul}|^2$; the different powers come from what $|A|^2$ is normalized against. In absorption, ε_2 is defined per unit *classical field intensity*: expressing $A_0 = cE_0/\omega$ through the measured field costs two inverse powers of ω (the -2 of Eq. (2.7)). In spontaneous emission there is no external field to normalize by; the reference object is *one photon per mode*, whose squared amplitude is fixed by the field quantization (2.3) at $2\pi\hbar c^2/\omega V$ — a *single* inverse power of ω , the zero-point amplitude of the mode. The remaining ω -dependence of (6.2) is carried openly by the mode count $\rho(\omega) \propto \omega^2$. In short: same vertex, different yardstick — absorption is measured against E_0^2 (two powers), emission against the vacuum fluctuation of one mode (one power).

6.3 The interband PL spectrum and the Kirchhoff check

The emission weight is not a difference but a *product*: the upper level must be filled and the lower empty, $f(E)[1 - f(E - \hbar\omega)]$. Summing (6.2) over pairs (spin 2, N half-neighborhoods, same $D_{l \rightarrow u}$ and windows as in absorption):

$$\boxed{R(\hbar\omega) = \rho(\omega) \frac{4\pi^2 e^2 \hbar |P|^2}{3m^2 \omega} 2N \int dE f(E)[1 - f(E - \hbar\omega)] D_{l \rightarrow u}(E, \hbar\omega)} \quad (6.3)$$

photons per unit time, volume and photon energy. In equilibrium the Fermi identity $[1 - f]/f = e^{(E - \mu)/k_B T}$, applied at both levels, collapses the product (detailed balance; only the *transition* energy survives):

$$f(E)[1 - f(E - \hbar\omega)] = n_B(\hbar\omega) [f(E - \hbar\omega) - f(E)], \quad n_B(x) = \frac{1}{e^{x/k_B T} - 1}, \quad (6.4)$$

and since $[f(E - \hbar\omega) - f(E)]$ is Rosei's $[1 - f(E)]$ in the same deep- d -band limit used throughout (Sec. 2.5), every electronic factor of (6.3) maps onto (11) and the constants collapse *exactly*:

$$\boxed{R(\hbar\omega) = \frac{\omega^3}{\pi^2 \hbar c^3} \varepsilon_2(\omega) n_B(\hbar\omega) = \frac{\omega \rho(\omega)}{\hbar} \varepsilon_2(\omega) n_B(\hbar\omega)} \quad (6.5)$$

— the van Roosbroeck–Shockley relation; per thermal photon, emission over absorption is $e^{-\hbar\omega/k_B T}$: Kirchhoff. That (9) and (6.3), derived independently, satisfy (6.5) with no leftover constant is the strongest global check that every prefactor in this document is right.

Out of equilibrium the collapse (6.4) fails — it used one shared (μ, T) at both levels — and the ratio $n_{\text{eff}}(E, \hbar\omega) = f(E)[1 - f(E - \hbar\omega)]/[f(E - \hbar\omega) - f(E)]$ retains genuine E -dependence: the emission line shape parts company with ε_2 . That deviation *is* the non-equilibrium PL signal this thesis is after; reproducing $n_{\text{eff}} \rightarrow n_B$ under f_{FD} is a unit test, *taking* it is throwing the signal away. In a structured environment, $\rho(\omega) \rightarrow \rho(\mathbf{r}, \omega)$ in (6.3).

7 Numerics: evaluate in $k_{\parallel}$, not in E

Groundwork for the implementation. The $1/\sqrt{}$ edges of (11) would force adaptive quadrature or endpoint-weighted rules — but the divergence is a pure coordinate artifact, manufactured in Sec. 3.4 when the smooth $\mathbf{k}$ -space geometry was re-parametrized by final energy. One exact change of variable inside the same 1D integral undoes it. Take $k \equiv k_{\parallel}$ itself. (Where did $k_{\perp}$ go? Nowhere — on the transition surface it is not an independent variable. With $\hbar\omega$ fixed, the constraint $\mathcal{E}_{lu} = \hbar\omega$ of (3.2) slaves the perpendicular coordinate, $k_{\perp}^2 = (\hbar\omega - \mathcal{E}_g - \bar{B}k_{\parallel}^2)/\bar{A}$, exactly as u was expressed through v in (3.5); substituting it into $E = \hbar\omega_u(\mathbf{k})$ of (3.1) leaves E a function of $k_{\parallel}$ alone, Eq. (7.2) below — $k_{\perp}$ is eliminated by the δ -constraint, not set to zero.) Then E is affine in k^2 , so $dE = \mp(\mu_{i\perp} \hbar^2 / \mathcal{F}_i^2) k dk$ while $D_{l \rightarrow u} \propto 1/k$ — the factors of k cancel identically,

$$D_{l \rightarrow u} dE = \mp \frac{\mu_{i\perp}}{4\pi^2 \hbar^2} dk_{\parallel}, \quad (7.1)$$

i.e. $k_{\parallel}$ is (up to constants) the cumulative count of the EDJDOS: integrating in $k_{\parallel}$ *is* integrating in E with the known weight $D_{l \rightarrow u}$ absorbed exactly into the node placement. Eq. (7) flattens to

$$\boxed{\mathcal{J}_i(\hbar\omega, T) = \frac{\mu_{i\perp}}{4\pi^2 \hbar^2} \int_{k_1^{(i)}}^{k_2^{(i)}} [1 - f(E_i(k), T)] dk} \quad \frac{1}{\mu_{i\perp}} \equiv \frac{1}{m_{u\perp}} + \frac{1}{m_{l\perp}}, \quad (12)$$

$$E_i(k) = \underbrace{\hbar\omega_{i_6^-} + \frac{\mu_{i\perp}}{m_{u\perp}} (\hbar\omega - \mathcal{E}_g^i)}_{k_{\parallel}=0 \text{ intercept}} \mp \frac{\mu_{i\perp} \hbar^2}{2\mathcal{F}_i^2} k^2 \quad (- \text{ at } X, + \text{ at } L), \quad (7.2)$$

valid on both sides of the X gap — all the branch logic lives in the limits. These are the $u = 0/v = 0$ endpoints of Sec. 4, written out since an implementation touches them first. At X (Rosei’s branch):

$$k_1^X = \begin{cases} 0, & \hbar\omega \geq \mathcal{E}_g^X, \\ \frac{1}{\hbar} \left[\frac{2(\mathcal{E}_g^X - \hbar\omega) m_{u\parallel} m_{l\parallel}}{m_{l\parallel} - m_{u\parallel}} \right]^{1/2}, & \hbar\omega < \mathcal{E}_g^X, \end{cases} \quad k_2^X = k_{\parallel}^X(E_{\text{floor}}, \hbar\omega) \text{ from (6')}, \quad (7.3)$$

with $E_{\text{floor}} = -20k_B T$ in equilibrium; for a pumped distribution take $E_{\text{floor}} = -(\hbar\omega_{\text{pump}} + 20k_B T)$ — generosity is cheap, since below the true floor the integrand is ≈ 0 . At L both limits are geometric:

$$k_1^L = 0, \quad k_2^L = \frac{1}{\hbar} \sqrt{2\mu_{L\parallel} (\hbar\omega - \mathcal{E}_g^L)}, \quad \frac{1}{\mu_{L\parallel}} \equiv \frac{1}{m_{u\parallel}} + \frac{1}{m_{l\parallel}}. \quad (7.4)$$

No singularity survives: the integrand of (12) is a bounded, smooth occupation profile whose only feature is the Fermi step of width $\sim k_B T$. **Recipe** (composite Simpson on a uniform grid is entirely adequate):

1. $k = \text{linspace}(k_1(\hbar\omega), k_2(\hbar\omega), N)$, N odd;

2. integrand $1 - f(E_i(k))$ — Fermi–Dirac in equilibrium, interpolation of a tabulated non-equilibrium distribution otherwise (use a *monotone* interpolant, e.g. PCHIP, to avoid overshoot at the distribution’s kinks);
3. `simpson(y, x=k)`, times $\mu_{i\perp}/4\pi^2\hbar^2$; then into (9).

Measured accuracy (equilibrium, $\hbar\omega = 2.4$ eV, against a converged reference): X reaches 5×10^{-10} relative error at $N = 101$ and machine precision by $N = 201$; the L integrand in equilibrium is constant across the window (Remark 4.1), and even a hot 2000 K distribution gives 3×10^{-14} at $N = 101$. By contrast, uniform-grid Simpson in E on a singular edge stalls near 10^{-2} even at $N = 6401$ ($O(\sqrt{\hbar})$ decay). Energy space *done properly* — the substitution $t = \sqrt{E_{\text{edge}} - E}$, or Gauss–Jacobi with the $1/\sqrt{\cdot}$ weight — is bit-identical to (12), because $t \propto k_{\parallel}$ exactly: the geometry had already named the correct quadrature variable. Hence the division of labor: *derive and reason in E ; evaluate in $k_{\parallel}$* .

Practical points.

- *Resolve the step*: the only feature sits at $k_{\text{step}} = k(E=0)$, trivially located since $E_i(k)$ is affine in k^2 ; at low T either raise N or split the interval there and Simpson each panel.
- *Respect the kinks* of a pumped distribution: slope discontinuities at $E_F \pm \hbar\omega_{\text{pump}}$ are *distribution* features no variable change removes — put panel boundaries at their images $k(E_F \pm \hbar\omega_{\text{pump}})$.
- *Vectorize over $\hbar\omega$* : build $k = k_1[:, \text{None}] + (k_2 - k_1)[:, \text{None}] \xi[\text{None}, :]$ with ξ uniform on $[0, 1]$ and Simpson along the last axis — the whole spectrum in one call. Gauss–Legendre with 48–64 nodes does the same job with fewer points if node count matters.
- *Cache the geometry*: $k_{1,2}$, $E_i(k)$ and the grid depend on $(\hbar\omega)$ only — not on T or on the distribution — so precompute them once and sweep temperatures/pump powers cheaply (this is what makes fitting fast).
- *Locality guard*: nothing in (12) knows where the parabolic neighbourhood ends — compare $k_2(\hbar\omega)$ with $\pi/4a \simeq 0.19 \text{ \AA}^{-1}$ (Sec. 5). In equilibrium the X integrand dies of occupation before locality does; a pumped distribution re-weights exactly those deep-window nodes, and the generous floor pushes k_2 to $\sim 2 \times \pi/4a$. Report the fraction of (12) accrued at $k > \pi/4a$ and distrust spectra where it is not small.
- *Error control*: recompute at $2N$ and compare (one doubling).
- *Emission*: the PL integrand (6.3) swaps $[1 - f] \rightarrow f(E)[1 - f(E - \hbar\omega)]$ in the same quadrature — geometry, windows and nodes unchanged.

Built-in checks. (i) *Empty-window arithmetic*: wherever the occupation factor is $\equiv 1$ across the window, $\mathcal{J}_i = \frac{\mu_{i\perp}}{4\pi^2\hbar^2}(k_2 - k_1)$ exactly; at L with $T \rightarrow 0$ this is the textbook M_0 onset $\mathcal{J}_L = \frac{\mu_{L\perp}}{4\pi^2\hbar^2} \sqrt{(\hbar\omega - \mathcal{E}_g^L)/\bar{B}_L}$ in closed form. (ii) *Two-forms agreement*: (11) by adaptive quadrature (endpoint declared) and (12) by Simpson must agree to quadrature accuracy at every $(\hbar\omega, T)$ — a one-line unit test. (iii) *Kirchhoff*: with $f = f_{\text{FD}}$, confirm $n_{\text{eff}}(E, \hbar\omega) \rightarrow n_B(\hbar\omega)$ independent of E (Sec. 6).

A Conventions

Fixed once here, used silently throughout the document.

Units	Gaussian throughout (Rosei's system); SI conversions noted where useful.
Energy zero	Fermi level, $E_F = 0$; all electron energies E from E_F .
Bands	u = upper (conduction), l = lower (d); Rosei's subscripts.
Levels & masses	$\hbar\omega_{X_6^-}, \hbar\omega_{X_7^+}, \hbar\omega_{L_6^-}, \hbar\omega_{L^+}$ and $m_{u\perp}, m_{u\parallel}, m_{l\perp}, m_{l\parallel}$ are all <i>positive numbers</i> ; curvature directions and the side of E_F are carried only by explicit $\pm$ signs, never hidden in a symbol.
Band coefficients	$A_i = \hbar^2/2m_{i\perp}$, $B_i = \hbar^2/2m_{i\parallel}$, $i = u, l$; all $A_i, B_i > 0$.
Interband energy	$\mathcal{E}_{lu}(\mathbf{k}) \equiv \hbar\omega_u - \hbar\omega_l$; Rosei's $\Omega_{lu} = \mathcal{E}_{lu} - \hbar\omega$, the CEDS being $\Omega_{lu} = 0$.
Local axes	$k_{\parallel}$ along ΓX (resp. ΓL), $k_{\perp}$ radial in the perpendicular plane; $\mathbf{k}$ measured from the critical point.
Spin	carried explicitly as a factor 2, never absorbed silently.
Counting	N counts equivalent <i>half</i> -neighborhoods ($N_X = 6$ faces, $N_L = 8$), paired with the half-space $D_{l \rightarrow u}$ below — or a factor 2 goes missing.
Occupation	Rosei's $[1 - f(E, T)]$, $f(E) = [e^{E/k_B T} + 1]^{-1}$; see Sec. 2.5.
Matrix element	$ P ^2 \equiv \langle u \nabla l \rangle ^2$ (dimension length ⁻²), Rosei's convention; momentum form $ \mathbf{p}_{ul} ^2 = \hbar^2 P ^2$.
Photon	energy $\hbar\omega$, wavevector $\mathbf{q}$, polarization $\hat{\epsilon}_{\lambda}$.

B Dictionary of conventions across the repository

Document	upper band	lower band	$\bar{B}$	status
Rosei 1975; this doc (X)	$+A_u k_{\perp}^2 - B_u k_{\parallel}^2$	$-A_l k_{\perp}^2 - B_l k_{\parallel}^2$	$B_l - B_u$	canonical
<i>Equivalence</i> note (X)	$+A_u k_{\perp}^2 + B_u k_{\parallel}^2$	$-A_l k_{\perp}^2 + B_l k_{\parallel}^2$	$B_u - B_l$	same $\mathcal{D}$ (sign-robust)
A8 / A8.L	$+A_c u - B_c v$	$-A_v u - B_v v$	$B_v - B_c$	A8 ok ^(a) ; A8.L wrong at $L^{(b)}$
<code>interband_derivation_v2</code>	mixed	mixed	—	internally inconsistent ^(c)

^(a) A8's $\mathcal{E}_{\min}$ denominator ($B_v + B_c$) is the $s=+1$ formula; for its own X conventions it should be $\bar{B} = B_v - B_c$. ^(b) A8.L models the L conduction band as a saddle ($-B_c$); C&S give $m_{u\parallel}^L = +0.12$: it is a minimum, $s = +1$. ^(c) v2 asserts $\bar{B} > 0$ and a geometric sub-gap tail (requires $\bar{B} < 0$), and labels L as M_1 .

Normalizations. Earlier repo documents use the full-neighborhood, no- $(2\pi)^3$ kernel $K(E, \hbar\omega) = \int d^3k \delta\delta$ and the half-line measure $\mathcal{J}_{\text{eq}}$ of the *Equivalence* note; with $D_{l \rightarrow u}$ from (4'),

$$K = 2(2\pi)^3 D_{l \rightarrow u}, \quad \mathcal{J}_{\text{eq}} = \frac{1}{2}K, \quad \text{printed (4)} = \frac{2\mathcal{F}}{\hbar^2} \times (4'). \quad (\text{B.1})$$

Matrix elements: Rosei's $|P|^2$ is the squared *gradient* element, related to the SI momentum element by $|\mathbf{p}_{ul}|^2 = \hbar^2 |P|^2$. With these two conventions — (4') and gradient $|P|^2$ — Rosei's Eq. (9) is exact as printed; no residual constant needs to be hidden in $|P|^2$, and only the strengths $S = \mathcal{F}|P|^2$ of Eq. (10) are fixed by the data (the printed-(4) constant reshuffles $D_{l \rightarrow u}$, $\mathcal{F}$, $|P|^2$ among themselves without touching S).

C Dimensional audit (Gaussian)

$[e^2] = \text{erg cm}$, $[|P|^2] = \text{cm}^{-2}$, $[A_i] = [B_i] = \text{erg cm}^2$, $[\mathcal{D}] = \text{erg}^2 \text{cm}^4$, $[\mathcal{F}] = \text{g}$, $[D_{l \rightarrow u}] = \text{erg}^{-2} \text{cm}^{-3}$, $[\mathcal{J}] = \text{erg}^{-1} \text{cm}^{-3}$.

- (4'): $\mathcal{F}/\hbar^4 k_{\parallel} \sim \text{g}^2/(\text{erg}^4 \text{s}^4 \text{cm}^{-1}) = \text{erg}^{-2} \text{cm}^{-3}$ — states per volume per (energy)², as a double- δ measure must be. ✓
- (9): $\frac{\text{erg cm} \cdot \text{erg}^4 \text{s}^4}{\text{g}^2 \text{erg}^2} \cdot \text{cm}^{-2} \cdot \text{erg}^{-1} \text{cm}^{-3} = 1$ (using $\text{erg s}^2 = \text{g cm}^2$). ✓
- (12): $\mu_{\perp}/\hbar^2 \times [k] \sim \text{g cm}^{-1}/\text{erg}^2 \text{s}^2 = \text{erg}^{-1} \text{cm}^{-3} = [\mathcal{J}]$. ✓

- (6.1): $[\rho] = \text{s cm}^{-3}$. ✓
- (6.2): $\frac{\text{erg cm} \cdot \text{s}^{-1} \cdot \text{erg s} \cdot \text{cm}^{-2}}{\text{g}^2 \text{cm}^3 \text{s}^{-3}} = \text{erg}^2 \text{cm}^{-1} / (\text{g}^2 \text{cm}^3 \text{s}^{-3} \text{s}) = \text{s}^{-1}$. ✓
- (6.3): $[\rho] \cdot [\text{matter}] \cdot [\mathcal{J}] = \text{s cm}^{-3} \cdot \text{cm}^3 \text{s}^{-2} \cdot \text{erg}^{-1} \text{cm}^{-3} = \text{erg}^{-1} \text{cm}^{-3} \text{s}^{-1}$ — photons per time, volume and photon energy. ✓
- (6.5): $[\omega\rho/\hbar] = \text{erg}^{-1} \text{cm}^{-3} \text{s}^{-1}$, times dimensionless $\varepsilon_2 n_B$. ✓

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